logistic regression

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SETTING A WORKING DIRECTORY

```
setwd("C:/Users/ADMIN/OneDrive/Desktop/caleb makini")
```

loading libraries

```
library(tidyverse)
library(rio)
library(ggplot2)
library(psych)
library(ggthemes)
library(summarytools)
library(tidyr)
library(formatR)
library(sjmisc)
library(caret)
library(report)
```

loading data

```
data <- read.csv("diabetes.csv")
head(data)
```

```
  age hypertension heart_disease smoking_history  BMI HbA1c_level
1  34            0             0           never 24.5         5.2
2  58            1             0          former 28.1         6.1
3  42            0             0         current 31.4         7.8
4  65            1             1          former 26.7         5.9
5  29            0             0           never 22.1         4.9
6  71            1             0           never 29.3         6.8
  blood_glucose_level diabetes
1                  89        0
2                 112        0
3                 145        1
4                  98        0
5                  82        0
6                 132        1
```

general linear models(GLM)

logistic regression

- is a type of regression analysis used for prediction of outcome of a categorical dependant variable based on one or more predictor variables.
- it is used when the dependant variable is binary(0/1. yes/ no, true /false)
- it estimates the probability of the depenant variable being 1(or less) given the values of the independent variables
- The logistic function is used to model the relationship between the dependant and independent variables.
- the logistic function is defined as:

omit missing values

```
omit_miss <- na.omit(data)
head(omit_miss)
```

```
  age hypertension heart_disease smoking_history  BMI HbA1c_level
1  34            0             0           never 24.5         5.2
2  58            1             0          former 28.1         6.1
3  42            0             0         current 31.4         7.8
4  65            1             1          former 26.7         5.9
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  blood_glucose_level diabetes
1                  89        0
2                 112        0
3                 145        1
4                  98        0
5                  82        0
6                 132        1
```

check columns name

```
names(data)
```

```
[1] "age"                 "hypertension"        "heart_disease"
[4] "smoking_history"     "BMI"                 "HbA1c_level"
[7] "blood_glucose_level" "diabetes"
```

convert variables to factor from integer and from character

```
data$diabetes <- factor(data$diabetes, levels = c(0, 1), labels = c("N
o", "Yes"))
data$hypertension <- factor(data$hypertension, levels = c(0, 1), label
s = c("No",
    "Yes"))
data$heart_disease <- factor(data$heart_disease, levels = c(0, 1), lab
els = c("No",
    "Yes"))
data$smoking_history <- factor(data$smoking_history)
```

confirm for missing values

```
colSums(is.na(data))
```

```
                age        hypertension       heart_disease     smokin
g_history
                  0                   1                   1
0
                BMI         HbA1c_level blood_glucose_level
diabetes
                  1                   1                   1
1
```

check the frequency of diabetes case

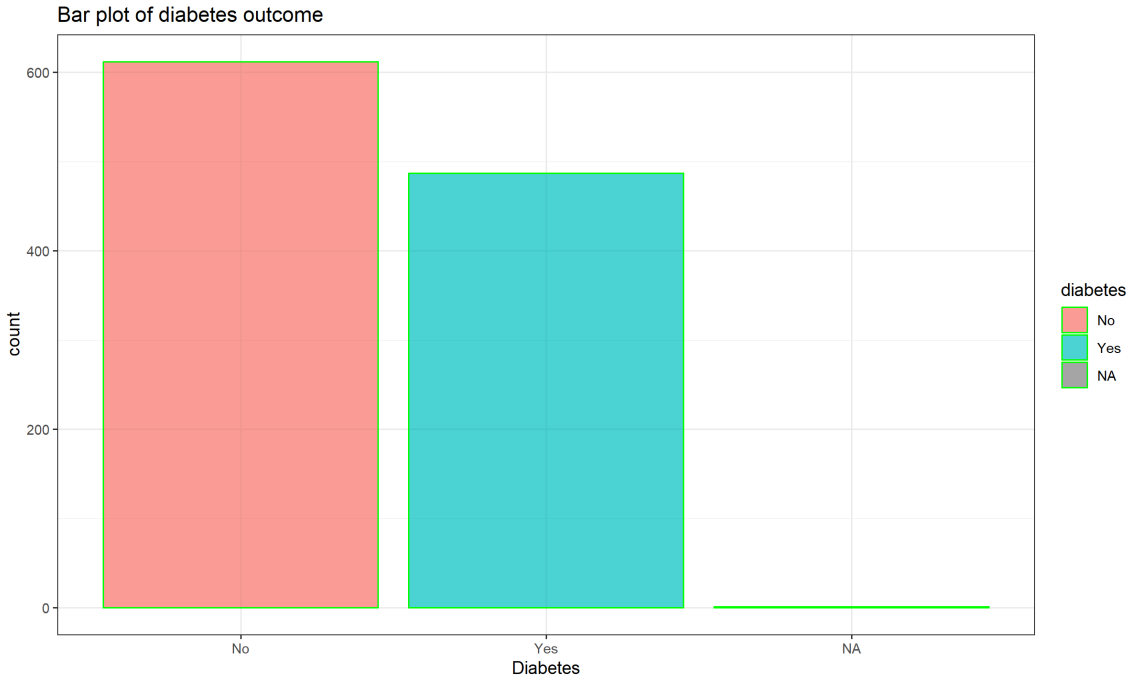
```
frq(data, diabetes)
```

```
diabetes <categorical>
# total N=1100 valid N=1099 mean=1.44 sd=0.50

Value |   N | Raw % | Valid % | Cum. %
--------------------------------------
No    | 612 | 55.64 |   55.69 |  55.69
Yes   | 487 | 44.27 |   44.31 | 100.00
<NA>  |   1 |  0.09 |    <NA> |   <NA>
```

visualize the frequency of diabates cases

```
ggplot(data, aes(x = diabetes, fill = diabetes)) + geom_bar(alpha = 0.
7, color = "green") +
    labs(title = "Bar plot of diabetes outcome", x = "Diabetes", y =
"count") + theme_bw()
```



partition the dataset into training and testing sets

```
set.seed(123)
trainIndex <- createDataPartition(data$diabetes, p = 0.8, list = FALS
E, times = 1)
train <- data[trainIndex, ]
test <- data[-trainIndex, ]
```

check the dimension of training and testing set

```
dim(train)
```

```
[1] 881   8
```

```
dim(test)
```

```
[1] 219   8
```

set the reference categories

```
data$smoking_history <- relevel(data$smoking_history, ref = "current")
data$hypertension <- relevel(data$hypertension, ref = "No")
data$heart_disease <- relevel(data$heart_disease, ref = "No")
```

fit a logistic regressin model

```
model <- glm(diabetes ~ ., data = train, family = binomial)
summary(model)
```

```

Call:
glm(formula = diabetes ~ ., family = binomial, data = train)

Coefficients:
                      Estimate Std. Error z value Pr(>|z|)
(Intercept)           -7825.19  149853.35  -0.052    0.958
age                      19.96     388.29   0.051    0.959
hypertensionYes       -2711.98   52705.70  -0.051    0.959
heart_diseaseYes        -379.95    7421.25  -0.051    0.959
smoking_historyformer    205.85    5810.35   0.035    0.972
smoking_historynever     388.37    8807.24   0.044    0.965
BMI                      135.62    2698.74   0.050    0.960
HbA1c_level           -1427.67   27410.57  -0.052    0.958
blood_glucose_level      115.17    2211.08   0.052    0.958

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 1.2086e+03  on 879  degrees of freedom
Residual deviance: 1.2407e-05  on 871  degrees of freedom
  (1 observation deleted due to missingness)
AIC: 18

Number of Fisher Scoring iterations: 25
```

Re-estimate the glm model by specifying the reference category for categorical variables

```
model1 <- glm(diabetes ~ age + hypertension + heart_disease + smoking_
history + BMI +
    HbA1c_level + blood_glucose_level, data = train, family = binomia
l)
summary(model1)
```

```
Call:
glm(formula = diabetes ~ age + hypertension + heart_disease +
    smoking_history + BMI + HbA1c_level + blood_glucose_level,
    family = binomial, data = train)

Coefficients:
                      Estimate Std. Error z value Pr(>|z|)
(Intercept)           -7825.19  149853.35  -0.052    0.958
age                      19.96     388.29   0.051    0.959
hypertensionYes       -2711.98   52705.70  -0.051    0.959
heart_diseaseYes       -379.95    7421.25  -0.051    0.959
smoking_historyformer   205.85    5810.35   0.035    0.972
smoking_historynever    388.37    8807.24   0.044    0.965
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(Dispersion parameter for binomial family taken to be 1)

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```

interpretation **

- unit in (positive integer Estimate) increases the probability of someone being diabetic
- unit in (negative integer Estimate) decreases the probability of someone being diabetic

report

```
report(model1)
```

```
We fitted a logistic model (estimated using ML) to predict diabetes wi
th age,
hypertension, heart_disease, smoking_history, BMI, HbA1c_level and
blood_glucose_level (formula: diabetes ~ age + hypertension + heart_di
sease +
smoking_history + BMI + HbA1c_level + blood_glucose_level). The mode
l's
explanatory power is substantial (Tjur's R2 = 1.00). The model's inter
cept,
corresponding to age = 0, hypertension = No, heart_disease = No,
smoking_history = , BMI = 0, HbA1c_level = 0 and blood_glucose_level =
0, is at
-7825.19 (95% CI [-14879.93, -770.46], p = 0.958). Within this model:

  - The effect of age is statistically non-significant and positive (b
eta =
19.96, 95% CI [-31.19, 260.90], p = 0.959; Std. beta = 235.87, 95% CI
[-368.49,
3082.77])
  - The effect of hypertension [Yes] is statistically non-significant
and
negative (beta = -2711.98, 95% CI [-5376.60, 90.71], p = 0.959; Std. b
eta =
-2711.98, 95% CI [-5376.60, 90.71])
  - The effect of heart disease [Yes] is statistically non-significant
and
negative (beta = -379.95, 95% CI [-5621.13, 637.94], p = 0.959; Std. b
eta =
-379.95, 95% CI [-5621.13, 637.94])
  - The effect of smoking history [former] is statistically non-signif
icant and
positive (beta = 205.85, 95% CI [-76.26, 461.10], p = 0.972; Std. beta
=
205.85, 95% CI [-76.26, 461.10])
  - The effect of smoking history [never] is statistically non-signifi
cant and
positive (beta = 388.37, 95% CI [-36.63, 795.90], p = 0.965; Std. beta
=
388.37, 95% CI [-36.63, 795.90])
  - The effect of BMI is statistically non-significant and positive (b
eta =
135.62, 95% CI [-261.05, 1880.43], p = 0.960; Std. beta = 510.42, 95%
CI
[-982.47, 7077.03])
  - The effect of HbA1c level is statistically non-significant and neg
ative (beta
= -1427.67, 95% CI [-18117.80, 2317.70], p = 0.958; Std. beta = -2566.
67, 95%
CI [-32572.28, 4166.77])
```

```
  - The effect of blood glucose level is statistically non-significant
and
positive (beta = 115.17, 95% CI [643.50, 669.67], p = 0.958; Std. beta
=
4106.29, 95% CI [22942.41, 23875.49])

Standardized parameters were obtained by fitting the model on a standa
rdized
version of the dataset. 95% Confidence Intervals (CIs) and p-values we
re
computed using a Wald z-distribution approximation.
```

model predictions

```
probabilities <- predict(model1, newdata = test, type = "response")
head(probabilities)
```

```
           1            2            4            7           15
16
2.220446e-16 2.220446e-16 2.220446e-16 1.000000e+00 2.220446e-16 1.000
000e+00
```

convert the probabilities to factors ‘Yes’ and ‘No’

```
predicted_class <- ifelse(probabilities > 0.5, "yes", "no")
predict_factor <- factor(predicted_class, levels = levels(test$diabete
s))
head(predicted_class)
```

```
    1     2     4     7    15    16
 "no"  "no"  "no" "yes"  "no" "yes"
```

confusion matrix

```
lg_cm <- confusionMatrix(predict_factor, reference = test$diabetes, po
sitive = "Yes")
lg_cm
```

```
Confusion Matrix and Statistics

          Reference
Prediction No Yes
       No   0   0
       Yes  0   0

               Accuracy : NaN
                 95% CI : (NA, NA)
    No Information Rate : NA
    P-Value [Acc > NIR] : NA

                  Kappa : NaN

 Mcnemar's Test P-Value : NA

            Sensitivity :  NA
            Specificity :  NA
         Pos Pred Value :  NA
         Neg Pred Value :  NA
             Prevalence : NaN
         Detection Rate : NaN
   Detection Prevalence : NaN
      Balanced Accuracy :  NA

       'Positive' Class : Yes
```

four fold confusion matrix plot

```
m <- lg_cm$byClass[c(1, 2, 5, 7, 11)]
m
```

```
      Sensitivity       Specificity         Precision                 F
1
               NA                NA                NA                 N
A
Balanced Accuracy
               NA
```

plotting confusion matrix plot

```
fourfoldplot(lg_cm$table, col = c(2, 3), main = "confusion matrix for
logistic regression model")
```

confusion matrix for logistic regression model

